

Introduction to medical surgical Nursing

Concepts of health and illness

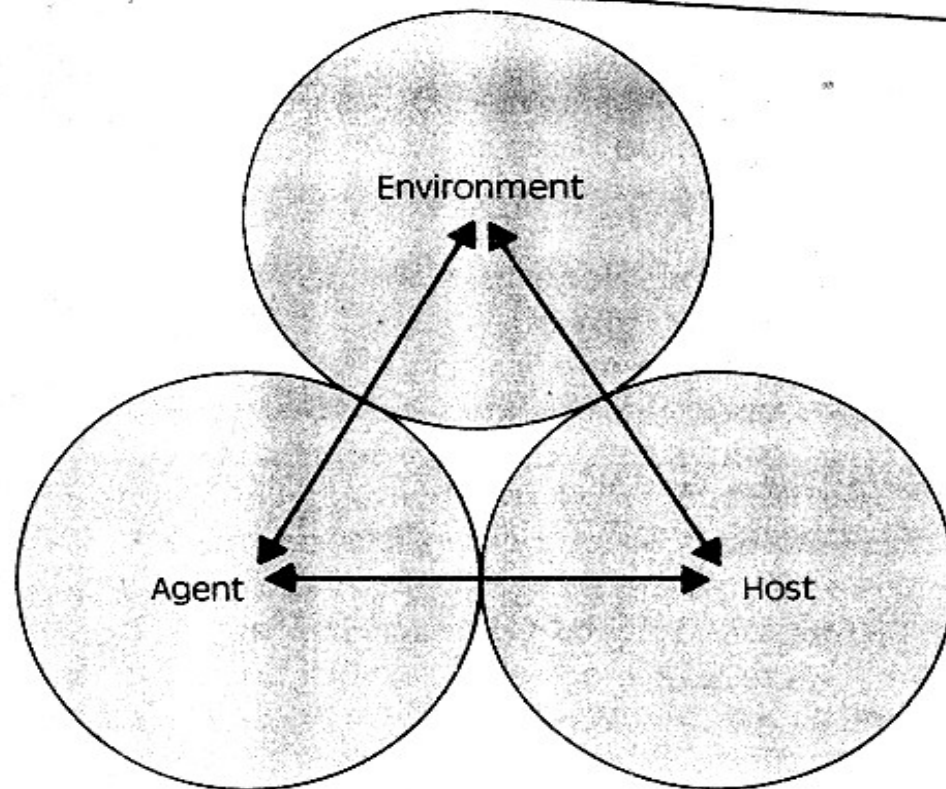
- The World Health Organization defines health as “a state of complete physical, mental and social well-being, not merely the absence of disease and infirmity.
- This definition considers the total persons state of health and wellness as essential component.

- Wellness is not only the absence of disease;
 - therefore, any definition of health should consider the different dimensions influencing health.
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- Health is a dynamic state in which the person is constantly adapting to changes in the internal and external environment.

Models of Health and illness

- **a. Host –agent-environment model**
- According to this model health and illness depends on interaction of host, agent and environmental factors.
- increases the possibility of illness. When the agent, host and environment variables are in equilibrium health is maintained.
- On the other hand when the balance is disrupted disease occurs.

Fig 1. Host –agent-environment model



Factors affecting Health and illness

- 1. **Physical dimension**-genetic make up, age, developmental level, race and sex
- 2. **Emotional dimension**-how the mind and body interact to affect to body function and to respond to body conditions also influence s health. Eg. long term stress affects the body systems, anxiety affects health
- 3. **Intellectual dimension**-encompasses cognitive abilities, educational background and past experiences.

- **4. Environmental dimensions**-the environment has many influences on health and illness. Housing sanitation, climate, pollution of air, food and water are aspects of the environmental dimension.
- **5. Sociocultural dimensions**- health practices are strongly influenced by a person's economic level, life style, family and culture.
- **6. Spiritual dimensions**- spiritual and religious beliefs and values are important components of how a person behaves in health and illness.

Nursing in wellness and holistic health care

- Nurses carry out wellness promotion activities on primary, secondary and tertiary levels

Preventing activities

Primary prevention: is a care directed toward health promotion and specific protection against illness. E.g. Immunization, family planning and health education

- **Secondary Prevention:** focuses on health maintenance for clients experiencing health problems on prevention of complication or disabilities.
E.g. Nursing care for hospitalized clients, early detection and treatment of health problems
- **Tertiary prevention:** is aimed at helping rehabilitate clients and restore them to a maximum level of functioning following an illness. E.g. teaching a diabetic client how to recognize and prevent complications

Introduction to nursing processes

- According to NANDA (**N**orth **A**merican **N**ursing **D**iagnostics **A**ssociation), the nursing process is defined as :
 - It is a **problem solving process** that nurses use in interacting with patients, their families or significant others in providing nursing care
- ☞ **A problem solving** is the basic skill of identifying a problem and taking steps to resolve it

Nursing process...

- In general, the nursing processes is a **continuous**, **scientific**, **systematic** client oriented, and goal oriented approach where the nurse and client work together:
 - ❖ To ensure quality care
 - ❖ To determine the need for nursing care
 - ❖ To plan and implement the care and evaluate the results
- Unlike the medical model, which focuses on treating the **disease**, the nursing processes is **holistic** .

The purpose of Nursing processes

Nursing processes is important for:

- 📄 Restoring, maintaining and promoting health
- 📄 Enabling individual or groups to manage their own health care to the best of their ability
- 📄 Providing nursing care

Characteristics of Nursing Process:

1. It is goal directed.
2. It is orderly and systematically organized.
3. It is dynamic and always changing.
4. It is interrelated and interactive.
5. It provides individual care.
6. It is patient centered.
7. It is practical for use over the life span.
8. It can be used for all setting.

Components of The Nursing process

- Assessment
- Diagnosis
- Planning
- Implementation
- Evaluation

A, Assessment

- Is a systematic and continuous collection and analysis of data or information about the client.
- ☐ The purpose of assessment is to identify whether the person is:
 - well,
 - has risk factors for problems, or
 - has actual problems

Assessment...

- Identify the following data as subjective or objective?
 - Headache
 - Bp =170/110 mmHg
 - Nausea
 - Abdominal pain
 - Skin lesion
 - Pain, fear , mood
 - Fever
 - Temperature =38 °C

Assessment...

Methods of data collection

☞ There are different methods of data collection and some of which includes:

- ❖ Observation
- ❖ Interview
- ❖ Laboratory testes
- ❖ Physical examination

Assessment...

Sources for the collecting data:

-  The patients (the best source) and their family
-  The physical examination
-  Health professionals
-  The patient's pervious and present records,
-  The laboratory reports

Assessment...

- The processes of gathering information include the following activities:

- 👍 Collecting data
- 👍 Recognizing significant data or validating data
- 👍 Recognizing patterns or clusters
- 👍 Identifying strengths and problems
- 👍 Prioritizing the data
- 👍 Reaching conclusions

B, The Nursing Diagnosis

- According to NANDA, the nursing diagnosis is defined, as "It is a clinical judgment about **individual, family or community** response to actual or potential health problems".
- It provides the basis for selection of nursing intervention to achieve outcome for which the nurse is accountable

The Nursing diagnosis ...

- Medicine clearly emphasizes the disease processes (HTN, pneumonia and DM).
- Nursing, however emphasizes on the person: the individual's response to his or her health and its focus became one of treating the whole person, not just the disease.

Medical Diagnosis vs Nursing

Diagnosis

Medical Diagnosis

Focuses on the illness, injury, or disease process.

Remains constant until a cure is effected.

Identifies conditions the health care practitioner is licensed and qualified to treat.

Nursing Diagnosis

Focuses on the responses to actual or potential health problems or life processes.

Changes as the client's response and/or the health problem changes.

Identifies situations in which the nurse is licensed and qualified to intervene.

Medical Diagnosis **vs** Nursing Diagnosis

E.g

➤ **Medical Dx** = pneumonia

➤ **Nursing Dx** =

Ineffective Airway clearance R/t tracheobronchial secretion.

The nursing diagnosis ...

Purpose of nursing diagnosis:

- ☐ Helps to identifies nursing priorities
- ☐ Directs nursing interventions to meet the clients high priority needs
- ☐ Provides communication between nursing professionals and the health care team
- ☐ Provides a base of evaluation

Components of Nursing Diagnosis

Nursing diagnoses has three components(PES)

1.Human response/Problem statement/diagnostic

label/definition = P

- It could be an actual problem or potential problem

2.Causative factor(Etiology)=E: which is the cause of the problem

3.Defining characters=S :are sign and symptoms that the pt manifests

TYPES OF NURSING DIAGNOSES

- 1. Actual nursing diagnosis → three part statement
 - 2. Possible nursing diagnosis
 - 3. Risk nursing diagnosis
- Two part statement
-
- 4. Wellness nursing diagnosis
 - 5. Syndrome nursing diagnosis
- One part statement

Actual nursing Dx

- ✓ It is client problem that is present at the time of the nursing assessment.
- ✓ It is considered as a 3 part statement i.e. it consists human response, causative factor and defining characters.

e.g

- ✓ Feeding self-care deficit **related to** right hemi paresis **as manifested by** inability to grasp utensils.
- ✓ Altered body °T **related to** the disease process **as evidenced by** body °T of 38 °C.

Possible nursing Dx

- Is a statement about a health problem that the client might have now, but the nurse doesn't yet have enough information to make an actual diagnosis. but a problem is only considered possible to occur.
- It is a two part statement, i.e. human response and causative factors will be included.

Examples:

- Possible nutritional deficit r/t nausea.
- Possible low self-esteem r/t loss job.

Risk Nursing diagnosis

- Is a clinical judgment that a problem does not exist, but special risk factors are present.
- Therefore no **S/S** are present, but the presence of **RISK FACTORS** indicates that a problem is only likely to develop unless nurse intervene or do something about it.
- Is a statement about a health problem that the client doesn't have yet, but it is at a higher than normal risk of developing in the near future.

Risk...

Examples:

- Risk for infection r/t surgical procedure.
- Risk for Constipation r/t inactivity and insufficient fluid intake
- Risk for infection r/t compromised immune system.

Wellness nursing diagnosis

- Clinical judgment about an individual, family and community in transition from a specific level of wellness to a higher level of wellness
- It is a one part statement ,i.e. only human response is described

E.g

- Effective therapeutic regimen
- Readiness for enhanced spiritual well-being

Syndrome nursing diagnosis

- It comprises a cluster of problems

E.g -: Rape-trauma syndrome.

Disuse Syndrome,

Post-Trauma Syndrome, and

Impaired Environmental Interpretation Syndrome

C. Planning

- This is the 3rd step of the nursing process which describes the development of goals to prevent, reduce or eliminate problems
- It also helps to identify nursing interventions that will assist clients in meeting the goals
- This is the step in which nurse will determine how to give nursing care in an organized, individualized, goal-directed manner

Planning...

- Planning involves several steps; that are important for preparing plan of nursing care, **these steps are**
 - ✍ Establishing or setting priorities to the nursing diagnosis and collaborative problems
 - ✍ Defining or establishing **goals** and **outcomes**
 - ✍ Determining specific nursing interventions
 - ✍ Recording the plan of care

Planning...

Establishing or setting priorities

- ❖ The nurse prioritizes the clients multiple problems by ranking first the diagnosis that are related to the client's most important, serious or immediate needs

Planning...

Look these examples of multiple nursing diagnoses:

- Spiritual distress
- Infective breathing pattern
- Risk for injury

Then, after looking your nursing diagnosis ask yourself
"which problem is need immediate attention and
remember the framework that helps you for prioritizing
the problems; Abraham Maslow.

Establishing goals and expected outcomes

Goals:

- Is an educated guess.
- addresses directly the problem stated in nursing diagnosis.

Expected outcome:

- Is a measurable client behavior that indicates whether the person has achieved the expected benefit of nursing care.

Con....

Example:

- Nursing diagnosis: knowledge deficit regarding postoperative care at home.

Goals: client will state three postoperative risk before discharge.

Expected outcomes:

- 1. Client will indentify need to drink 2-3 liters of every day.
- 2. Client will name three signs of infections.
- 3. Client will demonstrate aseptic wound care.

Planning...

- Look these examples of short and long-term outcomes respectively
 - ❖ The client will walk for 20 min longer each day of the first 3 postoperative day
 - ❖ The client will ambulate with a walker by the end of 3 weeks

Planning...

- Therefore, expected outcomes are the most important part of the care planning process, which are client oriented, specific, reasonable and measurable

- **The expected outcomes are:**
 - ✓ Used to measure to what extent progress toward resolving the problem has been made.
 - ✓ It is the basis for evaluating the effectiveness of the nursing diagnosis.
 - ✓ Helps in deciding whether additional nursing care is needed or whether the plan of care needs to be revised.

D,Implementation

- the 4 th step of the nursing process is called implementation.
- It is the action phase of the nursing process in which nursing care is provided.

Example of implementation

- Infective airway clearance related to physiologic effects of pneumonia as evidenced by increased sputum, coughing, abnormal breath sounds, tachypnea, and dyspnea

Implementation

- Administering supplemental humidified oxygen via nasal cannula at the prescribed flow rate
- Positioning the patient
- Assessing vital signs and respiratory status

Example of implementation...

- Begin intravenous (IV) fluid
- Instruct client in coughing
- Gradually increase client's activity level, assessing client out of bed to the chair
- Continue monitoring vital signs and respiratory status every 4 hours or as indicated,

E, Evaluation

- The final step of the nursing process, which is used to measuring the effectiveness of assessing, diagnosing, planning and implementing
- It also allow the nurse to determine the patients response to the nursing interventions and the extent to which the objectives have been achieved
- During evaluation, nurses compare the actual outcomes to the expected outcomes.
- Because this process enables the nurse to revise the expected outcomes

Eleven functional health patterns:

1. Health Perception and Health Management Pattern
- 2. Activity and Exercise Pattern**
- 3. Nutrition and Metabolism Pattern**
4. Elimination Pattern
- 5. Sleep and rest Pattern**
6. Cognition and Perception Pattern
7. Self-perception and Self-concept Pattern
8. Roles and Relationship Pattern
9. Coping and stress tolerance Pattern
10. Sexuality and Reproduction Pattern
11. Values and Belief Pattern

Activity and Exercise Pattern

➤ **Three areas may assessed:**

a. Mobility and self care

b. Respiratory function

c. Cardiovascular functions

a. Mobility and self care

Subjective data :

- Daily living activity-bathing, toileting, grooming, feeding.
- Simple motor activities-sitting, standing, walking, opening door.
- Home maintenance skills – cooking, shopping, house keeping (bed making, washing dishes, etc)
- Any restriction of activity or exercise
- Any thing that the patient is not doing any longer

➤ **Objective data:**

- Musculoskeletal assessment
- Motor examination(gait, posture, balance, coordination, abnormal movement, and body position)

Activity and Exercise...

b. Respiration function

➤ **Subjective data :**

- Risk factors for lung diseases such as smoking, exposure to pollutants, etc.
- Sign and symptoms of respiratory dysfunction such as cough, sputum production, chest pain, etc.
- Any impairment of functioning
- Medications

➤ **Objective data :**

- Respiratory pattern assessment
- Assessment of lung and thorax.

Activity and Exercise...

c. Cardiovascular functioning

➤ Subjective data :

- Risk factors for cardiovascular diseases such as family history, smoking, elevated cholesterol, hypertension, etc.
- Signs of cardiovascular dysfunction such fainting, palpitation, dizziness.
- Impact of cardiovascular problem on function
- Medications

➤ Objective data :

- Cardiovascular assessment

NUTRITION AND METABOLISM PATTERN

- This reflects how well the body is able to ingest, digest and metabolize food, use it to maintain tissue integrity and fluid and electrolyte balance, and to fight infection.
- **Subjective data:**
 - Normal food and fluid intake
 - Alteration in normal eating pattern including dietary restrictions and patient response to it.

NUTRITION AND METABOLISM...

➤ **Objective data:**

- General physical survey including weight, height and body mass index.
- Mouth examination(buccal mucosa, teeth, lips, gum tongue)
- Abdominal examination

SLEEP AND REST PATTERN:

➤ **Subjective Data:**

- Normal hours sleep per day
- Nap during day(when and how)
- Problems with obtaining adequate sleep and rest
- Any measure taken to induce sleep including medication and aids to reduce sleep.

➤ **Objective data:**

- Frequent yawning
- Decreased attention span
- Dark circles or puffiness around the eyes
- Continual dozing during the day

Bio-psychosocial concepts related to health

- **Stress, coping and adaptations**
- **Stress** is the body's physiological reaction to any stimulus
- Any situation, event, or agent that threatens a person's security is a **stressor**.

- A stressor can be physical (such as a laceration), or physiological (e.g., hypertension).
- The individual's *perception* of the stressor greatly determines whether the outcome is positive or negative.
- Any event can be stressful, depending on the person's interpretation of that event.

- **Adaptation** is an ongoing process by which individuals adjust to stressors in order to achieve **homeostasis** (equilibrium between physiological, psychological, sociocultural, intellectual, and spiritual needs).
- Adaptation is a holistic response that involves all dimensions of an individual.

- Individuals, as holistic beings, seek to maintain a *steady state* (another term for homeostasis) in all dimensions of
- life: physiological, psychological, cognitive, social, and spiritual.
- Wellness is an adaptive state; that is, the well person is one who is coping effectively with stressors to maintain a high level of well-being.
- The nurse's goal is to identify and support the client's positive adaptive responses.

factors contributing to stress

Physiological

- Trauma
- Illness
- Poor nutrition
- Sleep disturbances
- Hunger
- Discomfort
- Pain

Psychological

- Worry
- Fear
- Anger
- Happiness
- Thoughts
- Perceptions
- Interpretation of events

Sociocultural

- Job loss or promotion
- Changes in interpersonal relationships
- Interpersonal conflict
- Living conditions

Pain management

- Pain is an unpleasant sensory and emotional experience associated
- with actual or potential tissue damage
- It is the most common reason for seeking health care.
- It occurs with many disorders.
- It disables and distresses more people than any single disease.

- Since nurses spend more time with the patient in pain than do other health care providers,
- nurses need to understand the pathophysiology of pain,
- Thus, they must have the knowledge and skills to
- assess pain, to implement pain relief strategies, and to evaluate the effectiveness of these strategies.

- **The Fifth Vital Sign**
- Pain management is considered such an important part of care that the American Pain Society coined the phrase “Pain: The 5th Vital Sign”
- to emphasize its significance and to increase the awareness among health care professionals of the importance of effective pain management.
- Calling pain the fifth vital sign suggests that the assessment of pain should be as automatic as taking a patient’s blood pressure and pulse.

- The nurse collaborates with other health care professionals while administering most pain relief interventions,
- evaluating their effectiveness, and serving as patient advocate when the intervention is ineffective.
- In addition, the nurse serves as an educator to the patient and family,
- teaching them to manage the pain relief regimen themselves when appropriate.

Source

Source

- Nociceptive – sensory
- Neuropathic – from nerves
- Psychogenic – mental
- Area :to which it is referred or Localized
- Duration
 - Acute
 - Chronic

- Nociceptive
 - Cutaneous
 - Somatic
 - Visceral
- Neuropathic: is a collection of disorders that occurs when nerves of the peripheral nervous system (the part of the nervous system outside of the brain and spinal cord) are damaged.

Types of Pain

- Physical — cause of pain can be identified
- Psychogenic — cause of pain cannot be identified
- Referred — pain is perceived in an area distant from its point of origin

Duration of Pain

- Acute
 - Rapid in onset, varies in intensity and duration
 - Protective in nature
- Chronic
 - May be limited, intermittent, or persistent
 - Lasts for 6 months or longer
 - Periods of remission or exacerbation are common

Pain Responses

- Physiologic – automatic (involuntary) responses
- Behavioral – change in behavior
- Affective – emotional response

Assessing Pain

- **Severity**
- **Location**
- **Duration**
- **Characteristic**
- **New pain**
- **Aggravating/ relieving factors**
- **Medications/treatments presently using**

Diagnosing Pain

- **Type of pain**
- **Etiologic factors**
- **Behavioral, physiological, affective response**
- **Other factors affecting pain process**

Nursing Interventions

- **Establishing trusting nurse-patient relationship**
- **Initiating non-pharmacologic pain relief measures**
- **Considering ethical and legal responsibility to relieve pain**
- **Teaching patient about pain**

Nursing Plan

- **Remove or alter cause of pain**
- **Alter factors affecting pain tolerance**
- **Initiate non-pharmacologic relief measures**

Non-Drug Comfort Measures

- **Recreation or entertainment**
- **Music**
- **Relaxation**

Drug-Related Treatments

- **Analgesic administration**
 - **Non-opioid analgesics**
 - **Opioids or narcotic analgesics**

- Before administering analgesic agents, the nurse
- should assess the patient's pain status, including the intensity of
- current pain, changes in pain intensity after the previous dose of
- medication, and side effects of the medication.

- Three general categories of analgesic agents are opioids, NSAIDs, and local anesthetics. These agents work by different mechanisms.
- two or three types of agents simultaneously can maximize pain relief while minimizing the potentially toxic effects of any one agent.

- Administering analgesic medication on a time basis, rather than on the basis of the patient's report of pain,
- prevents the serum drug level from falling to sub therapeutic levels
- An example of this would be giving the patient the prescribed morphine or the prescribed NSAID (ibuprofen) every 4 hours rather than waiting until the patient complains of pain.

Opioid Analgesic Agents

- Eg Morphine, pethedine
- Opioids can be administered by various routes, including oral, intravenous,
- subcutaneous, and rectal routes.
- The goal of administering opioids is to relieve pain and improve quality of life
- therefore, the route of administration, dose, and frequency of administration are determined on an individual basis.

- Factors that are considered in determining the
- route, dose, and frequency of medication include the characteristics of the pain (eg, its expected duration and severity)
- the overall status of the patient, the patient's response to analgesic medications, and the patient's report of pain.
- Opioid analgesic agents given orally may provide a more consistent serum level than those given intramuscularly.

- RESPIRATORY DEPRESSION AND SEDATION
- Respiratory depression is the most serious adverse effect of opioid
- The risk of respiratory depression increases with age
- The patient receiving opioids by any route must be assessed frequently for changes in respiratory status.

- NAUSEA AND VOMITING
- Nausea and vomiting frequently occur with opioid use. Usually
- these effects occur some hours after the initial injection.
- The patient receiving an opioid should be assessed for nausea
- Adequate hydration and the administration of antiemetic
- agents may decrease the incidence.
- Opioid-induced nausea and vomiting often subside within a few days.

Introduction to onchology

- Cancer nursing practice covers all age groups.
- cancer nursing, also called **oncology nursing**.

Pathophysiology

- Cancer is a disease process that begins when an abnormal cell is transformed by the genetic mutation of the cellular DNA.

- This abnormal cell begins to proliferate abnormally.
- The cells acquire invasive characteristics, and changes occur in surrounding tissues.

- The cells infiltrate these tissues and gain access to lymph and blood vessels, which carry the cells to other areas of the body.
- This phenomenon is called **metastasis (cancer spread to other parts of the body)**.

Characteristics of Benign and Malignant Neoplasms

BENIGN

- Tumor does not infiltrate the surrounding tissues; usually encapsulated
- Rate of growth is usually slow
- Does not spread by metastasis

- Is usually a localized phenomenon that does not cause generalized effects
- Does not usually cause tissue damage unless its location interferes with blood flow
- Does not usually cause death unless its location interferes with vital functions

MALIGNANT

- Infiltrate and destroy the surrounding tissues
- Rate of growth the more anaplastic the tumor, the faster its growth
- Gains access to the blood and lymphatic channels and metastasizes to other areas of the body
- Often causes generalized effects, such as anemia, weakness, and weight loss
- produce substances that cause cell damage
- Usually causes death unless growth can be controlled

Cause

- **Viruses and Bacteria**
- **Viruses:**
- papillomavirus types are associated with cancer of the cervix.
- The hepatitis B virus is implicated in cancer of the liver.
- The human T-cell lymphotropic virus may be a cause of some lymphocytic leukemias and lymphomas.

- The human immunodeficiency virus (HIV) is associated with Kaposi's sarcoma.

Bacteria

- The bacterium *Helicobacter pylori* has been associated with an increased incidence of gastric malignancy

Palliative Care

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- The World Health Organization's (WHO's) define palliative care :
- Palliative care is the active total care of patients whose disease is not responsive to curative treatment.

- Palliative care is specialized medical care for people with serious illnesses.
- The goal of palliative care is achievement of the best quality of life for patients and their families.
- Many aspects of palliative care are also applicable earlier in the course of the illness.

- It is focused on providing patients with relief from the symptoms, pain, and stress of a serious illness whatever the prognosis.
- Control of pain, of other symptoms and of psychological, social and spiritual problems is paramount.

- Palliative care:
- provides relief from pain and other distressing symptoms
-
- Regards dying as a normal process;
- intends neither to hasten nor to postpone death;

Components of Palliative Care

- Effective symptom control
- Effective communication
- Rehabilitation – maximising independence
- Continuity of care
- Coordination of services
- Terminal care
- Support in bereavement

Purpose of palliative care:

- provides relief from pain, shortness of breath, nausea, and other distressing symptoms.
- Support life and regards dying as a normal process.
- Intends neither to hasten nor to postpone death;
- Integrates the psychological and spiritual aspects of patient care;

- To offer a support system to help patients live as actively as possible
- To offer a support system to help the family cope
- To enhance quality of life;

Management of clients with disruptions of the sensory system.

- **The eyeball(globe):**
- The eyeball, or globe, sits in a protective bony structure known as the orbit. Lined with muscle, connective and adipose tissues.
- **The eyelids-** composed of thin elastic skin that, protect the anterior portion of the eye.
- The eyelids contain multiple glands, including sebaceous, sweat, and accessory lacrimal glands, and they are lined with conjunctival material. ,
- The lids wash the cornea and conjunctiva with tears.

- Tears are vitally important to eye health. They are formed by the lacrimal gland.
- The conjunctiva, a mucous membrane, provides a barrier to the external environment and nourishes the eye.
- The sclera, commonly known as the white of the eye, is a dense, fibrous structure that comprises the posterior five sixths of the eye
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- **The sclera**- helps to maintain the shape of the eyeball and protects the intraocular contents from trauma.
- **The cornea** - is a transparent, avascular, domelike structure, forms the most anterior portion of the eyeball and is the main refracting surface of the eye.
- **The uvea** consists of the iris, the ciliary body, and the choroid.

- The iris, or colored part of the eye, is a highly vascularized, pigmented collection of fibers surrounding the pupil.
- The pupil is a space that dilates and constricts in response to light. Normal pupils are round and constrict symmetrically when a bright light shines on them.

- **Lens**-Directly behind the pupil and iris lies the lens,
- Lens is colorless and almost completely transparent.
- It is avascular and has no nerve or pain fibers.
- The lens enables focusing for near vision and refocusing for distance vision.
- The ability to focus and refocus is called **accommodation**.

- The retina contains the photoreceptor cells: **rods and cones.**
- Rods and cones are long, narrow cells shaped like rods or cones.
- The rods are mainly responsible for night vision or vision in low light.
- Whereas the cones provide the best vision for bright light, color vision, and fine detail.

Defnition of terms

- **1. Emmetropia:** Normal vision: rays of light coming from an object at a distance of 20 feet (6 m) or more are brought to focus on the retina by the lens.
- **2. Ametropia:** Abnormal vision.
- -A.**Myopia:** Nearsightedness: rays of light coming from an object at a distance of 20 feet or more are brought to a focus in front of the retina.

-B. **Hyperopia**: Farsightedness: rays of light coming from an object at a distance of 20 feet or more are brought to a focus in back of the retina.

- **3.Accommodation**: Focusing apparatus of the eye adjusts to objects at different distances by means of increasing the convexity of the lens (brought about by contraction of the ciliary muscles).

- 4. **Presbyopia**: The elasticity of the lens decreases with increasing age; an emmetropic person with presbyopia require prescription lenses to correct the problem.
- 5. **Astigmatism**: Uneven curvature of the cornea causing the patient to be unable to focus horizontal and vertical rays of light on the retina at the same time.
- 6. **Entropion** inward turning of the eyelid margin.
- 7. **Ectropion** outward turning of the eyelid margin.

Assessment

- **OCULAR EXAMINATION**
- After the patient's chief complaint or concern has been identified and the history has been obtained.
- visual acuity should be assessed.
- **Visual Acuity**
- Most health care providers are familiar with the standard Snellen chart.

- This chart is composed of a series of progressively smaller rows of letters and is used to test distance vision.
- The fraction 20/20 is considered the standard of normal vision.
- Normal eye can see the letters on the line designated as 20/20 from a distance of 20 feet (6m).
- The patient is positioned , usually 20 feet, from the chart and is asked to read the smallest line that he or she can see.

Taking an Ocular History

- Is visual acuity diminished?
- Does the patient experience blurred, double, or distorted vision?
- Is there pain; is it sharp or dull ?
- Is the discomfort an itching sensation or more of a foreign
body sensation?

- Are both eyes affected?
- Is there a history of discharge? If so, question color, consistency, odor.
- What is the duration of the problem?
- What makes the symptoms improve or worsen?
- Are there any systemic diseases? What medications are used in their treatment?

- Ophthalmoscopic Examination
- Direct ophthalmoscopy -uses a strong light reflected into the interior of the eye through an instrument called an ophthalmoscope.
- Helps to detect cataracts, corneal scars or inflammation.

Conditions of Eyelids and Conjunctiva

- **Blepharitis:**
- An inflammatory reaction of the eyelid margin **cause:**
- Bacteria (usually *Staphylococcus aureus*)
- **Signs and symptoms:**
- Redness, irritation, of the upper or lower lid, or both.
- **Treatment**
- Antibiotic ointment is prescribed 1 to 4 times per day to eyelid margin.

- **Hordeolum (stye)/Chalazion:**
- Refers to an inflammation or infection of the glands and follicles of the eyelid margin.
- External hordeolum involves the hair follicles of the eyelashes.
- chalazion is a granulomatous (chronic) infection of the meibomian glands.

Cause

- Bacteria, usually staphylococcus.
- **Signs and symptoms**
- Pain, redness, foreign body sensation, and a pustule may be present.
- **Treatment:**
- Antibiotic ointment
- warm soaks to help promote drainage
- Good hand washing and eyelid hygiene,
- In some cases, incision and drainage with local anesthetic may be necessary.

Conjunctivitis

- Inflammation or infection of the conjunctiva.
- **Cause**
- May be allergic,
- Bacterial
- Viral (adenovirus, herpes simplex).
- Irritative (Chemicals, wind, smoke, ultraviolet light)

Viral conjunctivitis

- **Signs and symptoms**
- Watery discharge
- Extreme tearing, redness, and foreign body sensation that can involve one or both eyes.
- Lid edema, ptosis, conjunctival **hyperemia** (ie, dilation of the conjunctival blood vessels)
- Viral conjunctivitis, although self-limited, tends to last longer than bacterial conjunctivitis.

Chlamydial conjunctivitis

- Chlamydial conjunctivitis includes trachoma
- Trachoma is an ancient disease and is the leading cause of preventable blindness in the world.
- **causes:**
- Trachoma is caused by *Chlamydia trachomatis*
- **Mode of transmission:**
- It is spread by direct contact or fomites, and the vectors can be insects such as flies .

Signs and symptoms of Trachoma

- The initial symptoms include red inflamed eyes, tearing, photophobia, ocular pain, purulent exudates, and lid edema.
- Initial ocular
- At the middle stage of the disease, there is:
- Trichiasis (turning inward of hair follicles) and entropion (inward turning of the eyelid margin). begin to develop.
- The lashes that are turned in rub against the cornea and, after prolonged irritation, cause corneal ulceration.

- Severe corneal ulceration can lead to perforation and blindness.
- **ALLERGIC CONJUNCTIVITIS**
- Immunologic or allergic conjunctivitis is a hypersensitivity reaction as a part of allergic rhinitis (hay fever),
- or it can be an independent allergic reaction.
- The patient usually has a history of an allergy to pollens and other environmental allergens.

- **Signs and symptoms:**
- There is extreme itching
- Epiphora (ie, excessive secretion of tears)
- Severe photophobia.
- Most affected individuals have a history of asthma or eczema.

Treatment

- Warm soaks (10 minutes four times per day) used when crusting and drainage present
- Cold compresses helpful for allergic and irritative causes.
- If topical antibiotic ordered, teach patient instillation technique.
- Urge good hand washing to prevent spread.
- Allergic conjunctivitis treated with topical or oral antihistamines.

- For trachoma:
- Treatment is usually broad-spectrum antibiotics.
- Administered topically and systemically. Surgical management includes:
- The correction of trichiasis to prevent conjunctival scarring.

DISORDERS OF THE CORNEA AND UVEAL TRACT

- The cornea is the outermost tissue that functions in vision.
- It must remain clear and smooth to admit light to the retina.
- The uveal tract is made up of:
 - -The iris, which controls pupil size.
 - -ciliary body, which secretes aqueous humor and controls accommodation.
 - - choroid layer, which provides vasculature to the anterior uveal tract.

- Disorders of the uveal tract may cause:
- - pupil changes
- - problems with accommodation.
- -Clouding of the anterior chamber or vitreous.

Keratitis

- Loss of epithelial layers of cornea.
- Cause:
 - due to some type of trauma
 - Contact with fingernail, tree branch, or other projectile, or overwearing contact lens.
- May lead to corneal ulceration and secondary infection into cornea (keratitis).
- which may lead to blindness.

- **Signs and symptoms**
- Pain
- Redness
- Foreign body sensation
- Photophobia
- Increased tearing
- Difficulty opening eye.

Treatment

- Treatment is urgent.
- Antibiotic ointment may be instilled and eye patched for 24 hours. Abrasion heals in 24 to 48 hours.
- Ulceration should be followed by an ophthalmologist.
- Make sure that patch is secure enough so patient cannot open eyelid.
- Review safety practices, such as wearing protective eye shields, not rubbing eyes, and washing hands frequently.

Uveitis

- Uveitis is an inflammation of the intraocular structures.
- It is classified by involved structures :
 - iritis) or iris.
 - Ciliary body (iridocyclitis)
 - Choroid (choroiditis)
 - Retina (retinitis)
 -

- Causes of uveitis are:
- Infections
- Immune-mediated disorders
- Trauma; or it may be idiopathic.
- **Signs and symptoms:**
- Onset is acute with deep eye pain
- Photophobia

- Conjunctival redness
- Small pupil that does not react
- Decreased visual acuity.
- **Treatment:**
- Urgent ophthalmology evaluation is needed.
- Inflammation is treated with a topical corticosteroid .
- Teach patient how to instill medications and adhere to dosing schedule to prevent permanent eye damage.

CATARACT

- Clouding or opacity of the crystalline lens that impairs vision.
- Cause:
- Senile cataract commonly occurs with aging
- Congenital cataract occurs at birth
- Traumatic cataract occurs after injury
- Additional risk factors for cataract formation include diabetes; ultraviolet light exposure; high-dose radiation; and drugs, such as corticosteroids, and some chemotherapy agents.

Clinical Manifestations

- Blurred or distorted vision.
- Gradual and painless loss of vision.
- Dark pupil may appear milky or white.
- Management
- Surgical removal of the lens is indicated.
- Cataract surgery is usually done under local anesthesia.
- Preoperative eyedrops produce decreased response to pain and lessened motor activity
- Oral medications may be given to reduce IOP.

- Appropriate eyeglasses or a contact lens to correct refraction after the healing process.
- Nursing Interventions
- Orient patient and explain procedures and care plan to decrease anxiety.
- Instruct patient not to touch eyes to decrease contamination.
- Administer preoperative eyedrops

ACUTE GLAUCOMA

- A condition in which an obstruction occurs at the anterior chamber angle.
- Glaucoma is associated with progressive visual field loss and eventual blindness if allowed to progress.
- **Cause:**
- Mechanical blockage of anterior chamber angle results in accumulation of aqueous humor (fluid).

- **Clinical Manifestations**
- Pain in and around eyes due to increased ocular pressure.
- Vision becomes cloudy and blurred.
- Pupil mid-dilated and fixed
- Nausea and vomiting may occur.

Management

- **Pharmacologic**
- Beta-adrenergic blockers may reduce production of aqueous humor or may facilitate outflow of aqueous humor.
- Hyperosmotic agents to reduce IOP by promoting diuresis.
- Surgery:
- Surgery is indicated if:

- IOP is not maintained within normal limits by medical regimen.
- There is progressive visual field loss with optic nerve damage.

- **Complications**

- Uncontrolled IOP that can lead to optic atrophy and total blindness.

Nursing Diagnoses

- Acute Pain related to increased IOP
- Fear related to pain and potential loss of vision
- Relieving Pain
- Nursing Interventions
- Administer opioids and other medications as directed.
- Reassure patient that, with reduction in IOP, pain and other signs and symptoms should subside.

- **Relieving Fear**
- **Nursing intervention.**
- Provide reassurance and calm to reduce anxiety and fear.
- Describe procedure to patient.

Anatomic and Physiologic Overview of the Ear

- The external ear
- The external ear - includes the auricle (pinna).
- The external auditory canal.
- The external ear is separated from the middle ear by a structure called the tympanic membrane (ie, eardrum).

- **Auricle**
- The auricle, attached to the side of the head by skin.
- The auricle collects the sound waves and directs vibrations into the external auditory canal.

- **External Auditory Canal**

- The external auditory canal is approximately 2.5 cm long.
- The skin of the canal contains hair, sebaceous glands, and ceruminous glands, which secrete a brown, wax like substance called cerumen (ie, ear wax).
- **ANATOMY OF THE MIDDLE EAR**
- The middle ear is connected by the eustachian tube to the nasopharynx.

- **Tympanic Membrane**
- The tympanic membrane protects the middle ear and conducts sound vibrations from the external canal to the ossicles.
- **Ossicles**
- The middle ear contains the three smallest bones (ie, ossicles) of the body: malleus, incus, and stapes.

- **ANATOMY OF THE INNER EAR**
- The organs for hearing (ie, cochlea)
- Balance (ie, semicircular canals)

- **Membranous Labyrinth**
- The membranous labyrinth contains a fluid called endolymph.

Assessment

- The external ear is examined by inspection and direct palpation.
- Tympanic membrane is inspected with an otoscope.
- Evaluation of gross auditory acuity also is included in every physical examination.

EAR DISORDERS

- **HEARING LOSS**
- Two major types of hearing loss are conductive and sensorineural hearing loss.
- **Conductive loss:** is hearing loss due to an impairment of the external or middle ear.
- **Sensorineural hearing:** is loss due to disease of the inner ear or nerve pathways.

OTOSCOPIC EXAMINATION

- The examiner looks for any discharge, inflammation, or foreign body in the external auditory canal.
- The position and color of the membrane and any unusual markings or deviations from normal are documented.
- The presence of fluid, air bubbles, blood, or masses in the middle ear also are noted.

External Ear condition

- **CERUMEN IMPACTION**
- Cerumen normally accumulates in the external Ear. wax does not usually need to be removed.
- Impacted cerumen(dry wax) may result to hearing loss.

Management

- Cerumen can be removed by irrigation.
- Contraindication:
- A perforated eardrum or an inflamed external ear (ie, otitis externa).
- Instilling a few drops of warmed glycerin, mineral oil, or halfstrength hydrogen peroxide into the ear canal for 30 minutes can soften cerumen before its removal.

- **EXTERNAL OTITIS (OTITIS EXTERNA)**
- **External otitis, or otitis externa,** refers to an inflammation of the external auditory canal.
- **Causes:**
- Bacterial or fungal infections are most frequently encountered.
- **Predisposing factor:**
- water in the ear canal (ie, swimmer's ear)

- **Clinical Manifestations**
- The patient usually reports pain, discharge from the external auditory canal,
- Other symptoms may include pruritus and hearing loss or a feeling of fullness.
- On otoscopic examination, the ear canal is erythematous and edematous.
- Discharge may be yellow or green and foul smelling.
- In fungal infections, the hair like black spores may even be visible.

- **Medical Management**
- The principles of therapy are aimed at relieving the discomfort,
- reducing the swelling of the ear canal, and eradicating the infection.
- Patients may require analgesics for the first 48 to 92 hours.
- Medications usually combine antibiotic and corticosteroid agents.
- For fungal disorders, antifungal agents are prescribed.

- **Nursing Management**
- Nurses need to teach patients:
- Not to clean the external auditory canal with cotton-tipped applicators
- To avoid swimming, and not to allow water to enter the ear when shampooing or showering.

Conditions of the Middle Ear

ACUTE OTITIS MEDIA

- Acute otitis media is an inflammation and infection of the middle ear caused by the entrance of pathogenic organisms.
- Lasting less than 14 days.
- **Cause**
- Bacteria
- Bacteria can enter the middle ear through eustachian tube from contaminated secretions in the nasopharynx

- **Clinical Manifestations**
- The pain is relieved after spontaneous perforation of the tympanic membrane.
- Drainage from the ear, fever, and hearing loss.
- On otoscopic examination, the external auditory canal appears normal.
- The patient reports no pain with movement of the auricle.
- The tympanic membrane is erythematous and often bulging.

- **Medical Management**
- Early and appropriate broad-spectrum antibiotic Eg- Cotrimexazole
- Wicking 3 times a day
- **SURGICAL MANAGEMENT**
- An incision in the tympanic membrane is known as **myringotomy**, to relieve pressure and to drain serous or purulent fluid from the middle ear.

- **Complications of acute otitis media**
- Mastoid and other serious intracranial complications, such as meningitis or brain abscess.
- **CHRONIC OTITIS MEDIA**
- **Chronic otitis media** is the result of repeated episodes of acute otitis media
- Chronic infections of the middle ear damage the tympanic membrane, destroy the ossicles, and involve the mastoid.

- **Clinical Manifestations**
- Varying degrees of hearing loss
- Presence of a persistent or intermittent, foul-smelling odour.
- Pain is not usually experienced, except in cases of acute mastoiditis
- Otoscopic evaluation of the tympanic membrane may show a perforation.

- **Medical Management**
- Wicking 3 times a day
- Application of antibiotic powder is used to treat a purulent discharge.
- Systemic antibiotics are usually not prescribed except in cases of acute infection.

OTOSCLEROSIS

- **Otosclerosis** involves the stapes and is thought to result from the formation of new, abnormal spongy bone, with resulting fixation of the stapes.
- The efficient transmission of sound is prevented because the stapes cannot vibrate and carry the sound as conducted from the malleus and incus to the inner ear.
- Predisposing factor: hereditary

- **Clinical Manifestations**
- The condition can involve one or both ears and manifests as a progressive conductive hearing loss.
- The patient may or may not complain of tinnitus.
- Otoscopic examination usually reveals a normal tympanic membrane.
- The audiogram confirms conductive hearing loss.

- **Medical Management**
- There is no known nonsurgical treatment for otosclerosis.
- **SURGICAL MANAGEMENT**
- A stapedectomy- removing the stapes, and inserting a suitable prosthesis.
- Stapes surgery is very successful in improving hearing.
- Balance disturbance or true vertigo, can occur for a short time after stapedectomy.

Conditions of the Inner Ear

- **MÉNIÈRE'S DISEASE**
- Ménière's disease is an abnormal inner ear fluid balance caused by a malabsorption in the endolymphatic sac.
- **Clinical Manifestations**
- Progressive sensorineural hearing loss
- Tinnitus
- A feeling of pressure or fullness in the ear
- Vertigo, often accompanied by nausea and vomiting.

Medical Management

- A low-sodium diet. The amount of sodium is one of many factors that regulate the balance of fluid within the body.
- PHARMACOLOGIC THERAPY
- Pharmacologic therapy for Ménière's disease consists of antihistamines which suppress the vestibular system.
- Tranquilizers such as diazepam (Valium) may be used in acute instances to control vertigo.

- Antiemetics such as promethazine (Phenergan) suppositories help control the nausea and vomiting and the vertigo because of their antihistamine effect.
- Diuretic therapy (eg, hydrochlorothiazide) sometimes relieves symptoms by lowering the pressure in the endolymphatic system.
- Intake of foods containing potassium (eg, bananas, tomatoes, oranges) is necessary if the patient takes a diuretic that causes potassium loss.

- **Labyrinthitis, an inflammation of the inner ear, can be bacterial or viral in origin.**
- Cause
- Bacterial labyrinthitis usually occurs as a complication of otitis media.
- The most commonly identified viral causes are mumps, rubella, rubeola, and influenza.
- Viral illnesses of the upper respiratory tract, also cause labyrinthitis.

- **Clinical Manifestations**
- Labyrinthitis is characterized by a sudden onset of vertigo, usually with nausea and vomiting
- Various degrees of hearing loss, and possibly tinnitus.
- **Management**
- Treatment of bacterial labyrinthitis includes intravenous antibiotic therapy, fluid replacement, and administration of a vestibular suppressant, such as antiemetic medications. Treatment of viral labyrinthitis is symptomatic.

Anatomy and Physiology of Respiratory system

- The primary purpose of the respiratory system is gas exchange
- which involves the transfer of oxygen and carbon dioxide between the atmosphere and the blood.
- The upper respiratory tract includes the nose, pharynx, adenoids, tonsils, epiglottis, larynx, and trachea.

- The lower respiratory tract consists of the bronchi, bronchioles, alveolar ducts, and alveoli.
- In adults, a normal tidal volume (V_T), or volume of air exchanged with each breath, is about 500 ml.
- Ventilation involves inspiration (movement of air into the lungs) and expiration (movement of air out of the lungs).

- The respiratory defense mechanisms include:
Filtration of air
- The cough reflex
- Alveolar macrophages.

Pleura:

The lungs and wall of the thorax are lined with a serous membrane called the pleura.

The visceral pleura covers the lungs
The parietal pleura lines the thorax.

ASSESSMENT

- During nursing assessment, a cough should be evaluated by the quality of the cough and sputum.
- During physical examination, the nose, mouth, pharynx, neck, thorax, and lungs should be assessed and the respiratory rate, depth, and rhythm should be observed.
- When listening to the lung sounds, there are extra breath sounds that are abnormal and include crackles, rhonchi, wheezes, and pleural friction rub.

DIAGNOSTIC STUDIES

- A chest x-ray is the most commonly used test for assessment of the respiratory system.
- Bronchoscopy is a procedure in which the bronchi are visualized may be used for diagnostic purposes to obtain biopsy specimens and assess changes resulting from treatment.

Upper Respiratory Problems

- Problems of the upper respiratory tract include disorders of the:
 - Nose
 - Pharynx
 - Tonsils,
 - Epiglottis
 - Larynx, and trachea.

A deviated septum & Nasal polyps

- **A deviated septum** is a deflection of the normally straight nasal septum
- Nasal septum deviates from the midline and can cause a partial obstruction
- **Nasal polyps** are tissue growths usually due to prolonged inflammation

Deviated septum and nasal polyps

- **cause:** trauma to the nose or congenital disproportion.

- **Clinical manifestations/assessment**

- Stertorous respirations (snoring)
- Dyspnea
- Postnasal drip

- **Management**

- Medical management/nursing interventions
 - Pharmacological management
 - Corticosteroids, antihistamines, antibiotics, analgesics

○Nasal polypectomy

○Rhinoplasty- the surgical reconstruction of the nose, is performed for cosmetic reasons or to improve airway function when trauma or developmental deformities result in nasal obstruction.

Allergic rhinitis

- Allergic rhinitis is the reaction of the nasal mucosa to a specific allergen.
- The most important step in managing allergic rhinitis involves identifying and avoiding triggers of allergic reactions.
- Acute viral rhinitis (also known as the common cold or acute coryza):
- **cause** adenovirus

management of acute viral rhinitis

- If viral rhinitis is the cause, medications are given to relieve the symptoms. Corticosteroids may be required
- If symptoms suggest a bacterial infection, an antimicrobial agent will be used
- Rest, fluids, proper diet, antipyretics, and analgesics are recommended.

PHARMACOLOGIC THERAPY

- Medication therapy for allergic and nonallergic rhinitis focuses on symptom relief.
- **Antihistamines** are administered for sneezing, itching, and rhinorrhea.
- **Oral decongestant agents** are used for nasal obstruction.
- In addition, intranasal corticosteroids may be used for severe congestion, and ophthalmic agents are used to relieve irritation, itching, and redness of the eyes.

- **Acute rhinitis (common cold)**

- Etiology/pathophysiology

- Inflammation of the mucous membranes of the nose and accessory sinuses
 - Virus(es)

- Clinical manifestations/assessment

- Thin, serous nasal exudate
 - Productive cough
 - Sore throat
 - Fever

- Medical management/nursing interventions
 - Pharmacological management
 - Analgesic
 - Antipyretic
 - Expectorant
 - Antibiotic (if infection present)
 - No specific treatment
 - Encourage fluids

ACUTE SINUSITIS

- Acute sinusitis is an infection of the paranasal sinuses.
- Cause
- It frequently develops as a result of an upper respiratory infection, such as an unresolved viral or bacterial infection, or allergic rhinitis.
- Nasal congestion, caused by inflammation, edema, and transudation of fluid, leads to obstruction of the sinus cavities
- This provides an excellent medium for bacterial growth, namely *Streptococcus pneumoniae*,

Clinical Manifestations

- Symptoms of acute sinusitis may include facial pain or pressure over the affected sinus area
- Nasal obstruction, fatigue, purulent nasal discharge
- Fever, headache, ear pain and fullness, dental pain, cough, a decreased sense of smell, sore throat or facial congestion or fullness.

Diagnostic Findings

- There may be tenderness to palpation over the infected sinus area.
- **Complications**
- Acute sinusitis, if left untreated, may lead to severe and occasionally life-threatening complications such as meningitis, brain abscess, ischemic infarction, and osteomyelitis.

Medical Management

- First-line antibiotics include amoxicillin, trimethoprim/sulfamethoxazole (Bactrim), and erythromycin.
- Second-line antibiotics include cephalosporins and amoxicillin clavulanate (Augmentin).

Nursing Management

- The nurse instructs the patient about methods to promote drainage such as inhaling steam
- Increasing fluid intake, and applying local heat (hot wet packs).
- The nurse stresses the importance of following the recommended antibiotic regimen, because a consistent blood level of the medication is critical to treat the infection.

CHRONIC SINUSITIS

- Chronic sinusitis is an inflammation of the sinuses that persists for more than 3 weeks in an adult and 2 weeks in a child.
- **Clinical Manifestations**
- Impaired mucociliary clearance and ventilation
- Cough (the thick discharge constantly drips backward into the nasopharynx)
- Chronic headaches in the periorbital area, and facial pain.
- In addition, some patients experience a decrease in smell and taste and a fullness in the ears.

Medical Management

- Medical management of chronic sinusitis is almost the same as for acute sinusitis.
- The antimicrobial agents of choice include amoxicillin clavulanate (Augmentin) or ampicillin .
- Clarithromycin and third-generation cephalosporins may also be used.
- The course of treatment may be 3 to 4 weeks
- Decongestant agents, antihistamines, saline sprays, may also provide some symptom relief.

SURGICAL MANAGEMENT

- When standard medical therapy fails, surgery, may be indicated to correct structural deformities that obstruct the openings of the sinus.
- Excising and cauterizing nasal polyps, correcting a deviated septum, incising and draining the sinuses and removing tumors are some of the specific procedures performed.

- **Nursing Management**
- The nurse teaches the patient how to promote sinus drainage by increasing the environmental humidity (steam bath, hot shower)
- Increasing fluid intake, and applying local heat.
- The nurse also instructs the patient about the importance of following the medication regimen.

ACUTE PHARYNGITIS

- Acute pharyngitis is an inflammation or infection in the throat.

Cause

- Most cases of acute pharyngitis are viral infection.
- Group A beta-hemolytic streptococcus, the most common bacterial Organism

- **Complications**
- include sinusitis, otitis media, mastoiditis.
- In rare cases the infection may lead to bacteremia, pneumonia, meningitis, rheumatic fever, or nephritis.

- **Clinical Manifestations**
- The signs and symptoms of acute pharyngitis include:
- Swollen lymphoid follicles with white-exudate
- Tender cervical lymph nodes and no cough.
- Fever, malaise, and sore throat also may be present.

Medical Management

- Viral pharyngitis is treated with supportive measures since antibiotics will have no effect on the organism.
- Bacterial pharyngitis is treated with a variety of antimicrobial agents.
- PHARMACOLOGIC THERAPY
- If a bacterial cause is suggested, penicillin is usually the treatment of choice.

- For patients who are allergic to penicillin or have organisms that are resistant to erythromycin cephalosporins may be used.
- Antibiotics are administered for at least 10 days to eradicate the infection from the oropharynx.
- Severe sore throats can also be relieved by analgesic medications, as prescribed. For example, acetaminophen

- **Nursing Management**
- The nurse instructs the patient to stay in bed to rest.
- Used tissues should be disposed properly to prevent the spread of infection.

- Warm saline gargles or irrigations are used
Irrigating the throat properly is an effective means of reducing spasm in the pharyngeal muscles and relieving soreness of the throat.

CHRONIC PHARYNGITIS

- Chronic pharyngitis is a persistent inflammation of the pharynx.
- It is common in adults who , use their voice to excess, suffer from chronic cough, and habitually use alcohol and tobacco.

Clinical Manifestations

- Patients with chronic pharyngitis complain of a constant sense of irritation or fullness in the throat.
- Mucus that collects in the throat and can be expelled by coughing.
- Difficulty swallowing.

Medical Management

- Treatment of chronic pharyngitis is based on relieving symptoms,
- Avoiding exposure to irritants
- If there is a history of allergy, one of the antihistamine decongestant medications, is taken orally every 4 to 6 hours.
- Aspirin or acetaminophen is recommended for its antiinflammatory and analgesic properties.

Nursing Management

- The nurse instructs the patient to avoid contact with others until the fever subsides.
- Alcohol, tobacco, second-hand smoke, and exposure to cold are avoided.
- The patient may minimize exposure to pollutants
- The nurse encourages the patient to drink plenty of fluids.
- Gargling with warm saline solutions may relieve throat discomfort.
- Lozenges will keep the throat moistened.

TONSILLITIS

- The tonsils are composed of lymphatic tissue and are situated on each side of the oropharynx.
- They frequently serve as the site of acute infection.
- Cause
- Group A beta-streptococcus is the most common organism associated with tonsillitis.

Clinical Manifestations

- The symptoms of tonsillitis include:
- Sore throat
- Fever, and difficulty swallowing.
- Swollen cervical lymph nodes
- White exudates on the throat

Medical Management

- Tonsillectomy is usually performed for recurrent tonsillitis when medical treatment is unsuccessful and there is severe hypertrophy.
- Appropriate antibiotic therapy
- is initiated.
- The most common antimicrobial agent is oral penicillin, which is taken for 7 days.
- Amoxicillin and erythromycin are alternatives.

LARYNGITIS

- **Laryngitis**, is an inflammation of the larynx.
- **Cause**
- Almost always a virus
- Bacterial invasion may be secondary
- **Predisposing factor**
- voice over use or exposure to dust, chemicals, smoke, and other pollutants.

- **Clinical Manifestations**
- Signs of acute laryngitis include hoarseness or **aphonia** (complete loss of voice)
- Severe cough.
- Chronic laryngitis is marked by persistent hoarseness.

Medical Management

- Management of acute laryngitis includes resting the voice, avoiding smoking, resting, and inhaling steam.
- If the laryngitis is due to a bacterial organism or if it is severe, appropriate antibacterial therapy is instituted.
- The majority of patients recover with conservative treatment; however, laryngitis tends to be more severe
- in elderly patients and may be complicated by pneumonia.

- For chronic laryngitis, the treatment includes resting the voice, eliminating any primary respiratory tract infection
- Eliminating smoking, and avoiding second-hand smoke.
- Topical corticosteroids, inhalation, may also be used. These preparations have no systemic
- or long-lasting effects and may reduce local inflammatory reactions.

- **Nursing Management**
- The nurse instructs the patient to rest the voice and to maintain a well-humidified environment.
- If laryngeal secretions are present during acute episodes, a daily fluid intake of 3 L are suggested to thin secretions.

Acute bronchitis

- Acute bronchitis is an infection of the lower respiratory tract that is generally an acute sequela to an upper respiratory tract infection.
- **Etiology and pathophysiology**
- Primarily viral etiology, but may also arise from bacterial agents.
- Airways become inflamed and irritated with increased mucus production.

Clinical Manifestations

- Dyspnea, fever, tachypnea.
- Productive cough, clear to purulent sputum.
- Pleuritic chest pain.
- **Diagnostic Evaluation**
- Sputum for Gram stain, culture, and sensitivity tests may be obtained to determine presence of bacterial infection.

Management

- Antibiotic therapy for 7 to 10 days may be indicated for patients with bacterial infection and/or underlying lung disease.
- Hydration and humidification.
- Secretion clearance interventions (may include controlled cough, chest physical therapy).

- Bronchodilators for bronchospasm and related cough in patients with evidence of airflow limitation.
- Symptom management for fever, cough.

EPISTAXIS

- **EPISTAXIS** is a hemorrhage from the nose, referred to as **epistaxis**.
- **cause**
- Rupture of tiny, distended vessels in the mucous membrane of any area of the nose.
- There are a variety of causes associated with epistaxis, including trauma, infection, inhalation of illicit drugs, cardiovascular diseases, a foreign body in the nose.

- **Medical Management**
- Initial treatment may include applying direct pressure.
- The patient sits upright with the head tilted forward to prevent swallowing and aspiration of blood
- Pinch the soft outer portion of the nose for 5 or 10 minutes continuously.
- If this measure is unsuccessful, a Topical vasoconstrictors, such as adrenaline ,may be prescribed.

PNEUMONIA

- Is an acute inflammation of the lung parenchyma.
- Pneumonia can be classified according to the causative organism, such as bacteria, viruses, *Mycoplasma*, *fungi*, *parasites*, and *chemicals*.
- Aspiration pneumonia occurs from abnormal entry of secretions or substances into the lower airway.

- Opportunistic pneumonia presents in certain patients with altered immune responses who are highly susceptible to respiratory infections.
- Examples of pneumonia in the immunocompromised host are *Pneumocystis carinii* pneumonia (PCP),
- PCP has been associated with AIDS.

- Fungal pneumonias, and mycobacterium tuberculosis.
- Immunocompromised states occur with the use of corticosteroids or other immunosuppressive agents
- Chemotherapy, nutritional depletion, use of broad-spectrum antimicrobial agents, AIDS, and genetic immune disorders

Pathophysiology

- Pneumonia arises from normally present flora in a patient whose resistance has been altered, or it results from aspiration of flora present in the oropharynx.
- It may also result from blood borne organisms that enter the pulmonary circulation.
- Pneumonia often affects both ventilation and diffusion.

- An inflammatory reaction can occur in the alveoli, producing an exudate that interferes with the diffusion of oxygen and carbon dioxide.
- White blood cells, mostly neutrophils, also migrate into the alveoli and fill the normally air-containing spaces.
- Areas of the lung are not adequately ventilated because of secretions and mucosal edema that cause partial occlusion of the bronchi or alveoli.

Risk Factor

- Cigarette smoking, COPD)
- Immunosuppressed.
- Prolonged immobility.
- Depressed cough reflex (due to medications)
- Aspiration of foreign material into the lungs during a period of unconsciousness (head injury, anesthesia,)

Clinical Manifestations

- A sudden onset of shaking chills
- Fever (38.5° to 40.5°C)
- Pleuritic chest pain that is aggravated by deep breathing and coughing.
- The patient is severely ill, with marked tachypnea (25 to 45 breaths/min)
- Rapid pulse

Diagnostic Findings

- History (particularly of a recent respiratory tract infection)
- Physical examination, chest x-ray studies, blood culture (bloodstream invasion, called bacteremia, occurs frequently), and sputum examination.

Medical Management

- The treatment of pneumonia includes administration of the appropriate antibiotic as determined by the results of the Gram Stain.
- Recommendations for treatment of pneumonia include: erythromycin, a doxycycline, amoxicillin [Augmentin]), trimethoprim–sulfamethoxazole and third-generation cephalosporins (ceftriaxone).

Preventive Measure

- Encourage smoking cessation.
- Perform suctioning and chest physical therapy if indicated.
- Minimize risk for aspiration by proper positioning of patient.

- Encourage reduced or moderate alcohol intake.
- Promote frequent turning, early ambulation and mobilization,
- effective coughing, breathing exercises, and nutritious diet.

- Hydration is a necessary because fever and tachypnea may result in fluid losses.
- Antipyretics may be used to treat headache and fever
- A Warm, moist inhalations are helpful in relieving bronchial irritation.
- If hypoxemia develops, oxygen is administered.
- Pulse oximetry analysis is performed to determine the need for oxygen.

Complications

- **SHOCK AND RESPIRATORY FAILURE**
- These complications are encountered chiefly in patients who have received no specific treatment or inadequate or delayed treatment.
- **ATELECTASIS AND PLEURAL EFFUSION**
- **Atelectasis** (from obstruction of a bronchus by accumulated secretions)
- **Pleural effusions:** after the pleural effusion is detected on a chest x-ray, a thoracentesis may be performed to remove the fluid.

NURSING PROCESS

- **THE PATIENT WITH PNEUMONIA**
- **Assessment**
- The nurse should monitor the following:
 - Changes in temperature and pulse
 - Amount, odor, and color of secretions
 - Frequency and severity of cough
 - Degree of tachypnea or shortness of breath

Diagnosis

- NURSING DIAGNOSES
- Based on the assessment data, the patient's major nursing diagnoses may include:
- Ineffective airway clearance related to copious tracheobronchial secretions
- Activity intolerance related to impaired respiratory function

- Risk for deficient fluid volume related to fever and dyspnea
- Imbalanced nutrition: less than body requirements
- Deficient knowledge about the treatment regimen and preventive health measures

- POTENTIAL COMPLICATIONS
- potential complications that may occur include:
 - Shock
 - Respiratory failure
 - Atelectasis
 - Pleural effusion
 - Confusion
 - Superinfection

Planning

- **Goals**
- The major goals for the patient may include:
- Improved airway patency
- **Nursing Interventions for improving airway patency**
- Remove secretions

- hydration (2 to 3 L/day) thins and loosens pulmonary secretions.
- Humidification may be used to loosen secretions and improve ventilation.

EXPECTED PATIENT OUTCOMES

- Expected patient outcomes may include:
- Exhibits no complications
- Has normal vital signs
- Reports productive cough
- Maintains or increases weight
- Complies with treatment protocol and prevention strategies

PLEURISY

- **Pathophysiology**
- Pleurisy (pleuritis) refers to inflammation of both layers of the pleurae (parietal and visceral).
- Cause:
- Pleurisy may develop in conjunction with pneumonia, TB, or after trauma to the chest, pulmonary infarction, or pulmonary embolism

Clinical Manifestations

- Pleuritic pain. Taking a deep breath, coughing, or sneezing, worsens the pain.
- Later, as pleural fluid develops, the pain decreases.
- **Diagnostic Findings**
- Diagnostic tests may include chest x-rays, sputum examinations,
- Thoracentesis to obtain a specimen of pleural fluid for examination.

- **Medical Management**
- Monitor for signs and symptoms of pleural effusion, such as shortness of breath, pain.
- Prescribed analgesics and topical applications of heat or cold provide symptomatic relief. Indomethacin (Indocin), a nonsteroidal anti-inflammatory drug (NSAID).

Nursing Management

- The nurse also can teach the patient to use the hands or a pillow to splint the rib cage while coughing.

PLEURAL EFFUSION

- Pleural effusion, a collection of fluid in the pleural space, usually secondary to other diseases.
- Normally, the pleural space contains a small amount of fluid (5 to 15 mL),
- which acts as a lubricant that allows the pleural surfaces to move without friction.
- **In pleural effusion**, an abnormal volume of fluid collects in the pleural space, causing shortness of breath.

Clinical Manifestations

- A large pleural effusion causes shortness of breath.
- When a small to moderate pleural effusion is present, dyspnea may be absent or only minimal.
- **Diagnostic Findings**
- Physical examination, chest x-ray, chest CT scan, and thoracentesis confirm the presence of fluid.

Medical Management

- Thoracentesis is performed to remove fluid, and to relieve dyspnea.
- Other treatments for malignant pleural effusions include surgical pleurectomy

EMPYEMA

- **EMPYEMA:** is accumulation of fluid in the pleural cavity.
- Cause:
- Most empyemas occur as complications of bacterial pneumonia or lung abscess.
- Other causes include penetrating chest trauma, nonbacterial infections, or after thoracic surgery .

Clinical Manifestations

- Fever, night sweats, pleural pain,
- Cough, dyspnea, anorexia, weight loss.

Diagnostic Findings

- The diagnosis is established by a chest x-ray or chest CT scan.

Medical Management

- The objectives of treatment are to drain the pleural cavity and to achieve full expansion of the lung.
- The fluid is drained and appropriate antibiotics, in large doses, are prescribed based on the causative organism.

Pulmonary edema

- **Pulmonary edema** is defined as abnormal accumulation of fluid in the lung tissue and/or alveolar space.
- It is a severe, lifethreatening condition.
- **Cause:**
- Hypervolemia or a sudden increase in the intravascular pressure
- Right heart failure

Clinical Manifestations

- Respiratory distress, characterized by:
- Dyspnea, air hunger, and central cyanosis.
- May become confused or stuporous.
- **Assessment and Diagnostic Findings**
- Auscultation reveals crackles
- The patient may be tachycardic, the pulse oximetry values begin to fall

Medical Management

- Management focuses on correcting the underlying disorder.
- Vasodilators, or contractility medications may be given.
- If the problem is fluid overload, diuretics are given and the patient is placed on fluid restrictions.
- Oxygen is administered to correct the hypoxemia

Nursing management

- Nursing management of the patient with pulmonary edema includes
- assisting with administration of oxygen if respiratory failure occurs.
- The nurse also administers medications (ie, morphine, vasodilators) as prescribed

Cor pulmonale

- **Cor pulmonale** is a condition in which the right ventricle of the heart enlarges (with or without right-sided heart failure).
- **Cause:**
- The most frequent cause is severe COPD, in which changes in the airway and retained secretions reduce alveolar ventilation.

Clinical Manifestations

- Symptoms of cor pulmonale are usually related to the underlying lung disease such as COPD.
- With right ventricular failure, the patient may develop increasing edema of the feet and legs
- Distended neck veins, an enlarged palpable liver, pleural effusion, and ascites.
- Headache, and confusion may occur as a result of increased levels of carbon dioxide (hypercapnia).
- Patients often complain of increasing shortness of breath, wheezing, cough, and fatigue.

- **Medical Management**
- Supplemental oxygen is administered to improve gas exchange
- Improved oxygen transport relieves the pulmonary hypertension that is causing the cor pulmonale.

- If the patient is in heart failure, hypoxemia and hypercapnia must be relieved to improve cardiac function and output.
- Bed rest, sodium restriction, and diuretic therapy also are instituted to reduce peripheral edema

Nursing management

- The nurse instructs the patient about the importance of close monitoring (fluid retention, weight gain, edema) and adherence to the therapeutic regimen.
- The nurse teaches the family to monitor for signs and symptoms of right ventricular failure and about emergency interventions and when to call for assistance.
- Encourage patient to stop smoking.

Pulmonary Embolism

- **Pulmonary embolism:** refers to the obstruction of the pulmonary artery or one of its branches by a thrombus (or thrombi) that originates somewhere in the venous system or in the right side of the heart.

Cause:

- Most commonly, it is due to a blood clot or thrombus.
- **Emboli:** air, fat, and septic.

- Most thrombi originate in the deep veins of the legs.
- A venous thrombosis can result from slowing of blood flow (stasis)
- When a thrombus completely or partially obstructs a pulmonary artery or its branches, the alveolar dead space is increased.
- Thus, gas exchange is impaired or absent in this area.

- **Clinical Manifestations**
- Dyspnea is the most frequent symptom
- Tachypnea (very rapid respiratory rate) is the most frequent sign .
- Chest pain is common

Risk Factors for Pulmonary Embolus

- Venous Stasis (slowing of blood flow in veins)
- Prolonged immobilization (especially postoperative)
- Prolonged periods of sitting
- Varicose veins
- Spinal cord injury

Predisposing Conditions

- Advanced age
- Obesity
- Pregnancy
- Oral contraceptive use
- **Prevention**
- Active leg exercises to avoid venous stasis, early ambulation,

Medical Management

- The treatment of Pulmonary Embolism may include a variety of modalities:
- General measures to improve respiratory and vascular status
- Emergency management
- Anticoagulation therapy
- Surgical intervention

- Emergency management consists of the following:
- Nasal oxygen is administered immediately to relieve hypoxemia,
- respiratory distress, and central cyanosis.
- Intravenous infusion lines are started to establish routes for medications or fluids that will be needed.

PHARMACOLOGIC THERAPY

- Anticoagulant therapy (heparin, warfarin sodium) for managing acute deep vein thrombosis
- Heparin is used to prevent recurrence of emboli but has no effect on emboli that are already present.
- A surgical embolectomy is rarely performed.

Nursing Management

- The nurse encourages ambulation and active and passive leg exercises to prevent venous stasis in patients on bed rest.
- The nurse also advises the patient not to sit or lie in bed for prolonged periods, not to cross the legs, and not to wear constricting clothing.

LUNG CANCER

- **LUNG CANCER:** is an Abnormal cell growth, and eventually a malignant cell.
- The pulmonary epithelium undergoes malignant transformation from normal epithelium to eventual invasive carcinoma.

- **Risk Factors**
- Risk factors for lung cancer, include:
- Tobacco smoke, second-hand (passive) smoke
- Environmental and occupational exposures, gender, genetics, and dietary deficits.

Clinical Manifestations

- The most frequent symptom of lung cancer is:
- Cough.
- The cough starts as a dry, persistent cough, without sputum production.
- When obstruction of airways occurs, the cough may become productive due to infection.

- Wheezing is noted (occurs when a bronchus becomes partially obstructed by the tumor).
- Patients also may report dyspnea.
- Hemoptysis or bloodtinged sputum.
- Pain also is a late manifestation and may be related to metastasis to the bone.
- Dysphagia, head and neck edema.
- weakness, anorexia, and weight loss also

- **Diagnostic Findings**
- A chest x-ray is performed to search for pulmonary density.
- CT scans of the chest are used to identify small nodules not visualized on the chest x-ray
- Sputum cytology is rarely used to make a diagnosis of lung cancer

Treatment

- In general, treatment may involve surgery, radiation therapy, or chemotherapy—or a combination of these.
- SURGICAL MANAGEMENT
- The most common surgical procedure for a small, apparently curable tumor of the lung is lobectomy (removal of a lobe of the lung).
- In some cases, an entire lung may be removed (pneumonectomy)

- **RADIATION THERAPY**
- Radiation also may be used to reduce the size of a tumor, to relieve the pressure of the tumor on vital structures.
- **CHEMOTHERAPY**
- Chemotherapy is used to alter tumor growth patterns, to treat patients with distant metastases or small cell cancer of the lung.

- Chemotherapy may provide relief, especially of pain, but it does not usually cure the disease.
- PALLIATIVE THERAPY
- Palliative therapy may include radiation therapy to shrink the tumor to provide pain relief

Nursing Management

- Nursing care includes strategies to ensure relief of pain and discomfort and to prevent complications.
- MANAGING SYMPTOMS
- RELIEVING BREATHING PROBLEMS
- Airway clearance techniques. This may be include: Deep-breathing exercises, chest physiotherapy, cough, and suctioning.

Chest Trauma

- **Blunt chest trauma** results from sudden compression or positive pressure to the chest wall.
- **Cause**
- Motor vehicle crashes, falls, and bicycle crashes are the most common causes of blunt chest trauma.
- **Penetrating trauma** occurs when a foreign object penetrates the chest wall.
- causes of penetrating chest trauma include gunshot wounds .

- Injuries to the chest are often life-threatening and result in one or more of the following pathologic mechanisms:
- Hypoxemia from disruption of the airway; injury to the lung parenchyma, or rib cage
- Hypovolemia from massive fluid loss from the great vessels
- Cardiac failure from increased intrathoracic pressure

Medical Management

- The goals of treatment is to initiate aggressive resuscitation.
- An airway is immediately established with oxygen support.
- Re-establishing fluid volume.
- Draining or removing any air or fluid from the thorax to relieve pneumothorax, or hemothorax

PENETRATING TRAUMA:

- **GUNSHOT AND STAB WOUNDS**
- Gunshot and stab wounds are the most common types of penetrating chest trauma.
- Knives cause most stab wounds.
- Pneumothorax, hemothorax, with severe and continuing hemorrhage, can occur from any small Wound.
- Gunshot wounds to the chest may damage the chest organs and great vessels.

Medical Management

- After the status of the peripheral pulses is assessed, a large-bore intravenous line is inserted.
- A chest tube is inserted into the pleural space in most patients with penetrating wounds of the chest to achieve rapid and continuing re-expansion of the lungs.

- The insertion of the chest tube frequently results in a complete evacuation of the blood and air.
- If the patient has a penetrating wound of the heart and great vessels, the esophagus, or the tracheobronchial tree, surgical intervention is required.

Chronic Obstructive Pulmonary Disease

- **Chronic obstructive pulmonary disease (COPD)** is a disease state characterized by airflow limitation that is not fully reversible.
- COPD may include diseases that cause airflow obstruction (eg, emphysema, chronic bronchitis) , bronchiectasis, and asthma

- **Chronic Bronchitis**
- Chronic **bronchitis**, a disease of the airways, is defined as the presence of cough and sputum production for at least 3 months in each of 2 consecutive years.
- Bronchitis is narrowed and has impaired air flow due to multiple mechanisms: inflammation, excess mucus production,
- and potential smooth muscle constriction (bronchospasm).

- **Bronchiectasis:** chronic dilation of a bronchus or bronchi.
- **Emphysema:** a disease of the airways characterized by destruction of the walls of over distended alveoli.

- **Emphysema**
- In emphysema, impaired gas exchange (oxygen, carbon dioxide)
- results from destruction of the walls of overdistended alveoli.
- “Emphysema” is a pathological term that describes an abnormal distention of the air spaces beyond the terminal bronchioles, with

- Destruction of the walls of the alveoli.
- As the walls of the alveoli are destroyed (a process accelerated by recurrent infections)
- Impaired oxygen diffusion, which leads to hypoxemia.
- In the later stages of the disease, carbon dioxide elimination is impaired,
- resulting in increased carbon dioxide tension in arterial blood (hypercapnia) and causing respiratory acidosis.

- Thus, right-sided heart failure (cor pulmonale) is one of the complications of emphysema.
- Congestion, edema, distended neck veins, or pain in the region of the liver suggests the development of cardiac failure.

Clinical Manifestations

- COPD is characterized by three primary symptoms: cough, sputum production, and dyspnea on exertion.
- Dyspnea may be severe and often interferes with the patient's activities.
- Weight loss is common because dyspnea interferes with eating.
- Often the patient cannot participate in even mild exercise because of dyspnea.

- The patient with COPD is at risk for respiratory infections, which in turn increase the risk for acute and chronic respiratory failure.
- In COPD patients, chronic hyperinflation leads to the “barrel chest”,
- This results from fixation of the ribs in the inspiratory position (due to hyperinflation) and from loss of lung elasticity.

Management of Client with COPD

- For patients with chronic obstructive pulmonary disease (COPD), physical activity is an important part of their quality of life.
- PHARMACOLOGIC THERAPY
- Bronchodilators relieve bronchospasm and reduce airway obstruction.

- **Corticosteroids.** Inhaled and systemic corticosteroids (oral or intravenous) may also be used in COPD but are used more frequently
- Oxygen therapy can be administered to prevent acute dyspnea.

- PULMONARY REHABILITATION
- Pulmonary rehabilitation for patients with COPD is well established
- and widely accepted as a means to alleviate symptoms

NURSING PROCESS:

For patient with COPD

- **Assessment**
- Assessment involves obtaining information about current symptoms as well as previous disease manifestations.
- **NURSING DIAGNOSES**
- Based on the assessment data, the patient's major nursing diagnoses may include the following:
- Ineffective airway clearance related to broncho constriction, increased mucus production, and ineffective cough.

- Ineffective breathing pattern related to shortness of breath, mucus, bronchoconstriction, and airway irritants
- Activity intolerance due to fatigue, ineffective breathing patterns, and hypoxemia
- Ineffective coping related to reduced socialization, anxiety, depression, and the inability to work

- POTENTIAL COMPLICATIONS
- Based on the assessment data, potential complications that may develop include:
 - Respiratory insufficiency or failure
 - Atelectasis
 - Pulmonary infection
 - Pneumothorax

Goals

- The major goals for the patient may include:
- improve gas exchange, airway clearance,
- Improve breathing pattern
- Improve activity tolerance, maximal self-management,
- improve coping ability, adherence to the therapeutic program and home care, and absence of complications.

Nursing Interventions

- The nurse should educate the patient regarding the hazards of smoking and cessation strategies and provide , counseling.
- Education is focused on rehabilitative therapies to promote independence in executing activities of daily living.

- The nurse instructs the patient to avoid extremes of heat and cold.
- Heat increases the body temperature, thereby raising oxygen requirements.
- Cold tends to promote bronchospasm.
- Air pollutants such as fumes, smoke, initiate bronchospasm.
- High altitudes aggravate hypoxemia.

- The nurse monitors pulse oximetry values to assess the patient's need for oxygen and administers supplemental oxygen as prescribed.

EXPECTED PATIENT OUTCOMES

- Demonstrates knowledge of hazards of smoking
- Demonstrates improved gas exchange
- Shows no signs of restlessness
- Has stable pulse oximetry
- Achieves maximal airway clearance
- Stops smoking

- Maintains adequate hydration
- Knows signs of early infection and is aware of how and when to report them if they occur
- Improves breathing pattern
- Practices and uses pursed-lip and diaphragmatic breathing

Bronchiectasis

- **Bronchiectasis** is a chronic, irreversible dilation of the bronchi and bronchioles.
- Cause:
- Airway obstruction
- Diffuse airway injury
- Pulmonary infections and obstruction of the bronchus
- Genetic disorders such as cystic fibrosis

- A person may be predisposed to bronchiectasis as a result of recurrent respiratory infections .
- **Clinical Manifestations**
- Symptoms of bronchiectasis include:
- Chronic cough and the production of purulent sputum.
- Hemoptysis.
- Clubbing of the fingers because of respiratory insufficiency.
- Repeated episodes of pulmonary infection.

- **Assessment and Diagnostic Findings**
- A definite sign is offered by the prolonged history of productive cough.
- Sputum consistently negative for tubercle bacilli. The diagnosis is
- A computed tomography (CT) scan, which demonstrates presence of bronchial dilation.

Medical Management

- Postural drainage is part of all treatment plans because draining the bronchiectatic areas by gravity
- reduces the amount of secretions and the degree of infection.
- Chest physiotherapy, including percussion and postural drainage, is important in secretion management.
- Infection is controlled with antimicrobial therapy based on the results of sensitivity tests.

Asthma

- **Asthma** is a chronic inflammatory disease of the airways.
- predisposing factor for asthma.
- Exposure to airway irritants or allergens

- **Pathophysiology**
- The inflammation leads to obstruction from the following:
- Swelling of the membranes that line the airways (mucosal edema)
- Contraction of the bronchial smooth muscle (bronchospasm)
- Increased mucus production

- **Symptoms**
- Wheezing
- Cough
- Dyspnea
- Chest tightness

- **Complications**
- Complications of asthma may include
- Status asthmaticus
- Respiratory failure
- Pneumonia
- Atelectasis.

Medical Management

- **PHARMACOLOGIC THERAPY**
- Corticosteroids eg prednsolone
- Aminophylline
- Theophylline
- Adrenaline